

Orhan Kaplan, PhD

Senior Data Scientist & Research Specialist | PhD | Specialist in Causal Inference, Machine Learning, & Quantitative UX

Instructor at Nizip Vocational School, Gaziantep University

CONTACT

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EDUCATION

Ph.D.in Curriculum and Instruction, *University of Florida*, Gainesville, 2018

Dissertation: *Analyzing the Effects of Motivation, Expectations and Self-concept on Eighth-grade Students' Mathematics Achievement*

Major: Curriculum and Instruction, with a specialization in Mathematics Education

Committee: Thomasenia L. Adams (Chair), Aki Murata, Pavlo Antonenko, M. Kristina DePue

M.S.E. in Mathematics, *Arkansas State University*, Jonesboro,

2011 Major: Mathematics

Thesis: *Stone-Weierstrass Approximation Theorem, A constructive Approach*

Committee: Jie Miao (Advisor), Debra Ingram, Susanne Melescue

ELS English Course, *University of Saint Thomas*, Houston, 2009

Intensive English Program

B.S. in Elementary Mathematics Education, *Gazi University*, Ankara/Turkey, 2006

Major: Teaching Elementary School Mathematics

Minor: Teaching Elementary School Science

RESEARCH INTERESTS

Program and Impact Evaluations, Behavioral and Psychological Modeling, Predictive and Causal

APPOINTMENTS AND EMPLOYMENTS

Lecturer, April 2019 –

- Nizip Vocational School & Nizip Faculty of Education, Gaziantep University, Gaziantep/Turkey [Completing mandatory service due to liabilities tied to MEB study abroad scholarship]

Strategic Planning & Quality Development Specialist. , November 2018 – March 2019

- Research and Strategy Identification Coordination Team (ASKE), Batman Provincial Directorate of National Education, Batman/Turkey [Mandatory service due to liabilities of MEB study abroad scholarship]

Graduate Research Assistant, August 2017 – May 2018

- Higher Education Department at University of Florida, Gainesville, FL.
- Cleaning and managing secondary data using Stata
- Educational policy data screening and statistical analyses for causal inferences
- Literature review

Research Assistant (OPS), October 2016 - May 2017

- Higher Education Department at University of Florida, Gainesville, FL.
- Cleaning and managing secondary data using Stata
- Data screening and analyses

Mathematics Tutor, July 2012 – August 2012

- University Athletic Association, Strategy Tutoring Program, Gainesville, FL
- Tutoring undergraduate student-athletes

Mathematics Instructor, January 2012- May 2012

- Upward Bound Program at University of Florida, Gainesville, FL
- Instructed high school students in ACT-oriented mathematics. College preparatory program for students in poverty.

Mathematics & Science Teacher, September 2007-January 2008

- Sehit Ogretmenler Middle School, Batman/ Turkey

- Instructed students in middle school mathematics
- Mentored students in constructing a strategic education plan

Mathematics & Science Teacher, September 2006- February 2008

- İkiztepe Elementary & Middle School, Batman/ Turkey
- Instructed students in middle school mathematics and science
- Counseled students in constructing a strategic education plan
- Weekend mathematics course for rural students in poverty who prepared for the High School Entrance Exam

Mathematics Tutor, September 2005- February 2006

- Boyut Private Tutoring Center, Ankara/ Turkey
- Private Institution where students were prepared for the University Entrance Exam (OSS) in Turkey. Worked as a mathematics tutor.

PUBLICATIONS

BOOKS AND BOOK CHAPTERS

Kaplan, O., & Bozkurt, B. (2025). Immigrant education policies and practices in Sweden. In M. Şenel, E. Akcan, & B. Bozkurt (Eds.), *Comparative immigrant education policies and practices* (pp.369-389). Sakarya University Press.

Kaplan, O. (2023). Cebir Öğretimi [Teaching Algebra]. In F. Erdogan (Ed.), *Matematik ve fen bilimleri eğitiminde yeni yaklaşımlar 2023-II* [New approaches to mathematics and science education 2023-II] (pp. 277-294). Efe Akademi Publication.
<https://doi.org/10.59617/efepub2023>

Celik, A., **Kaplan, O.**, Kavun, S., Kavun, S., Kures, M.B., & Kucukdegirmenci, T. (2006). *Memorial Exercises*, 4th Grades, ATA Publishing House, Ankara, Turkey.

PEER-REVIEWED ARTICLES

Temurtas, A. & **Kaplan, O.** (In Review). Shared Structures Between Mathematics and Science Motivational Beliefs: A Canonical Investigation. *International Journal of Science and Mathematics Education*.

Kaplan, O. & Kacar, T. (In Review). Profiling Preservice Teachers Who are at Risk of Digital Technology Addiction Using CHAID Analysis. *Education and Science*.

Aydın Gurler, S. & **Kaplan, O.** (Accepted). The Mediating Role of Belief in the Importance of Art in the Relationship Between Attitudes toward STEAM and Creative Personality Traits. *Journal of History School*.

Aydın Gürler, S. & **Kaplan, O.** (2026). The effect of school bullying on science achievement and cognitive appraisals. *Problems of Education in the 21st Century*, 84(1), 7–26. <https://doi.org/10.33225/pec/26.84.07>

Kaplan, O. (2026). Cross-cultural control–value mechanisms on the detrimental effect of bullying on mathematics anxiety. *Behavioral Sciences*, 16(1), Article 3. <https://doi.org/10.3390/bs16010003>

Kaplan, O. (2025). Does algebra curriculum develop cognitively advanced critical minds? Comparative analysis of Türkiye and the U. S. A. mathematics textbooks. *Electronic Social Sciences Journal*, 24(3), 1297–1320. <https://doi.org/10.17755/esosder.1592018>

Kaplan, O., Kara, M., & Kaçar, T. (2025). The effect of digital technology integration on students' creative thinking skills: PISA evidence using propensity score matching. *Problems of Education in the 21st Century*, 83(3), 403–423. <https://doi.org/10.33225/pec/25.83.403>

- Kaplan, O.** (2024). Thematic modeling of pre-service mathematics teachers' reflections on CHATGPT use: appraisals, ethical challenges and aspirations. *Adiyaman University Journal of Educational Sciences* 14 (1), 21-35. <http://dx.doi.org/10.17984/adyuebd.1463794>
- Aydın Gürler, S. & **Kaplan, O.** (2023). Attitudes towards STEAM, critical thinking disposition and decision-making skills: Mediation and gender moderation. *International Journal of Contemporary Educational Research*, 10(1), 210-223. <https://doi.org/10.33200/ijcer.1272051>
- Kaplan, O.,** & Adams, T. L. (2023). How social expectations affect eighth-grade students' math achievement: An investigation of motivational drivers. *Journal of Innovative Research in Teacher Education*, 4(1), 92-112. <https://doi.org/10.29329/jirte.2023.531.6>
- Baydu, M. M., **Kaplan, O.,** & Bayar, A. (2013). Facing the influence of poverty on graduation rates in public high schools. *Procedia - Social and Behavioral Sciences*, 84, 233–237. <https://doi.org/10.1016/j.sbspro.2013.06.541>

PRESENTATIONS

- Kaplan, O.,** Kacar, T. & Simsek, N. (2025). The effect of bullying on epistemic emotions using Propensity Score Matching: Examining cross-national differences. The paper was presented at 2025 *European Conference on Educational Research*, Belgrade, Serbia.
- Kaplan, O.** (2024). Comparative Analysis of Connections in Algebra Learning Area in Secondary School Mathematics Textbooks: Associations with Algebraic Concepts and STEAM Disciplines. The paper was presented at 16. *Ulusal Matematik ve Fen Bilimleri Kongresi* [16th National Mathematics and Science Congress], Edirne, Turkey.
- Kaplan, O.** (2023). Öğretmen adaylarının matematik öğretiminde ders imecesi: Yansıtıcı

- düşünme becerileri ve gerekçelendirmede eksiklikler [Preservice teachers' lesson study in mathematics teaching: Deficiencies in reflective thinking skills and justification]. The paper was presented at *15. Ulusal Matematik ve Fen Bilimleri Kongresi* [15th National Mathematics and Science Congress], Kars, Turkey.
- Aydın Gürler, S. & **Kaplan, O.** (2022). Examination of primary school students' attitudes towards STEAM in terms of some variables. The paper was presented at *8. International Zeugma Conference on Scientific Research*, Gaziantep, Turkey.
- Kaplan, O.** & Alagoz, B. (2019). Syrian immigrants' integration and inclusion in Turkey's educational context: educational policy, practice, issues and recommendations. The paper was presented at *3rd International Symposium of Education and Values*, Istanbul, Turkey.
- Wagner, R., **Kaplan, O.**, & Elliott, K. (2018). NCTM's voice in the evolution of assessment, equity, and feminism from 1920's to the present. The paper was presented at *NCTM Annual Meeting & Exposition*, Washington, DC.
- Kaplan, O.** & Adams, T. L. (2017). Learning from the past, projecting the future: social justice from NCTM's lens. The paper was presented at *NCTM Regional Conference*, Orlando, FL.
- Kaplan, O.** (2017). The development of equity in education through NCTM's Lens. The poster was presented at *1st Annual Diversity Graduate Research Symposium*, Gainesville, FL.
- Kaplan, O.** (2015). Factors affecting mathematics teachers' perspective toward CCSSM. The poster presented at *NCTM Research Conference*, Boston, MA.
- Kaplan, O.** & Mitten, C. (2014). An exploratory study of mathematics teachers' incorporation of the Common Core State Standards for Mathematics. The poster was presented at the *9th Annual Conference of the Social Sciences*, Gainesville, FL.
- Gaulee, U., Lee, B., Ridgewell, N. B., Butcher, C., Koro-Ljunberg, M., Whitaker, D., Almazam, F., **Kaplan, O.**, & Watson, D. M. (2013). Constructivist study of graduate students' health care experiences. The paper was presented at *Florida Educational Research Association (FERA) 58th Annual Meeting*, Gainesville, FL.
- Kaplan, O.** (2013). Teachers' Readiness for Incorporating the Common Core State Standards for Mathematics (CCSS-M) Scale. The paper was presented at *60th*

Annual Southeastern Regional Association of Teacher Educators (SRATE) Conference, Sarasota, FL.

Kaplan, O., Adams, T. L., & Mitten, C. (2013). Barriers to efficiently incorporating Common Core State Standards for Mathematics (CCSS-M) in mathematics instruction: An exploratory study. The paper was presented at *60th annual Southeastern Regional Association of Teacher Educators (SRATE) Conference, Sarasota, FL.*

Baydu, M. M., Bayar, A., & **Kaplan, O.** (2012). Facing the influence of poverty on graduation rates in public high schools. The paper was presented at *3rd World Conference on Psychology, Counseling and Guidance, Izmir, Turkey.*

Kaplan, O. & Miao, J. (2011, March). Stone-Weierstrass Approximation Theorem, A Constructive Approach. Paper was presented at Symposium of Research & Scholarship, Arkansas State University, Jonesboro, AR

TECHNICAL PROJECTS

A Diagnostics Package for Multiple Imputation by Chained Equations (using PMM), June 2026 – July 2026

https://github.com/nalpako2027/mice_imputation_diagnostics_helper.git

This Github repository is an open-source Python implementation of Multiple Imputation by Chained Equations (MICE) with Predictive Mean Matching (PMM), designed for complex survey data. The package implements the imputation methodology described by Rubin (1987) and van Buuren (2018) and includes a comprehensive suite of diagnostics—such as Rubin's Rules pooling statistics, the Fraction of Missing Information (FMI), Relative Variance Increase (RVI), Relative Efficiency (RE), Von Hippel's recommended number of imputations, convergence trace plots, and before/after imputation population statistics. These capabilities were not available together in existing open-source Python implementations at the time of development.

Telco Customer Churn Analysis & Predictive Modeling Pipeline | Python, Github June, 2026 – June, 2026

https://github.com/nalpako2027/telco_customer_churn_prediction_simulation.git

This project contains an end-to-end data science and business analytics pipeline

designed to identify high-risk customer churn segments and evaluate the financial impact of proactive marketing interventions using the IBM Telco Customer Churn dataset using logistic regression and KNN classification. Features include custom data quality assertions, optimization of decision thresholds, multi-model ROC evaluation, and simulation of the financial ROI of customer retention strategies.

Sci-Fi Movie Recommendation System Using KNN Collaborative Filtering | Python, Github May, 2026 – May, 2026

https://github.com/nalpako2027/collaborative-filtering_movie-recommender

Engineered an end-to-end Sci-Fi content recommendation engine using a distance-based KNN Regression framework in Python. Optimized the data pipeline using Manhattan distance (L_1 norm) and a rigorous 75/25 train-test split validation to choose the best KNN.

GRANTS & FELLOWSHIPS

Researcher, *Step-by-Step to Independence Project*, November, 2024 - November, 2025

The project aimed to organize nature-based activities and walking events to support university students' physical and mental health in the fight against addictions. Within the scope of the project, six walking routes in the Gaziantep region were in the project.

Instructor, *Applications of AI in Mathematics Instruction* for the TUBITAK Project

222B158 'Technological and Innovative Content Development Applications for Mathematics Education', May 22nd - May 26th, 2023.

Studying Abroad Scholarship, sponsored by Ministry of National Education 2007

A comprehensive scholarship covering education for Master's and PhD degrees

AWARDS AND HONORS

Certificate of Appreciation, 2026

Appreciation for quality academic publications, and dedicated work and contributions to

faculty events (co-project manager in establishing a Mathematics Library in Nizip Faculty of Education).

Honorable Mention Award, 2014

For the poster presented at the 9th Annual Conference of the Social Science, Gainesville, FL

Certificate of Appreciation, 2012

For the service and commitment to the Upward Bound Program at the University of Florida as a geometry instructor

Certificate of Achievement, 2010

Turkish Educational Attaché, New York

LICENCES AND CERTIFICATION

HarvardX – Machine Learning and AI, 2026

Provided rigorous hands-on training in building and optimizing advanced predictive models, utilizing powerful ensemble learning techniques including bagging, random forests, and boosting algorithms—alongside cross-validation frameworks to solve complex classification and regression problems.

Machine Learning and AI Foundations: Causal Inference and Modeling, 2026

Certificate of Completion by LinkedIn Learning

Learning Git and Github, 2026

Certificate of Completion by LinkedIn Learning

HarvardX – Introduction to Data Science with Python, 2026

An online course covering statistical learning, model selection and cross-validation, visualization, and probabilistic prediction models using Python. Analysis technique included multiple regression, logistic regression, kNN regression and classification, Ridge & Lasso regressions.

Measurement and Evaluation Certificate, 2019

Measurement and Evaluation Seminars, Gaziantep University

Intensive English Program, 2009

Advanced Level Certification, ELS Language Centers

Professional Educator Certificate, 2007

Middle School Mathematics Teacher, Turkish National Ministry of Education Certified

PROFESSIONAL SERVICE

Member of the Organizing Committee, BİLTEK-IV 4th International Symposium on Current Developments in Science, Technology and Social Sciences, Gaziantep University, Nizip Faculty of Education, Nizip, Gaziantep, 2021

Program Committee Volunteer, NCTM's Annual Meeting & Exposition, Washington, DC, 2018. Contributions: Assisting the program committee on site; Greeting speakers and monitoring rooms; Monitoring technical issues

RESEARCH/TECHNICAL SKILLS

- ⇒ Full Proficiency in Academic English
- ⇒ Statistical Analysis software: SPSS, Stata, MPlus, Dedoose, R Program, Python
- ⇒ Python programming language
- ⇒ Data cleaning, data management and statistical analyses on large-scale datasets
- ⇒ Quantitative Data Analysis:
 - Descriptive statistics and visualization
 - Parametric and non-parametric tests
 - Rating scale design and analysis
 - Exploratory and confirmatory factor analyses
 - Other advanced statistical analyses i.e., regression, path analysis, structural equation modeling

- Statistical analysis for causal inferences: Regression-discontinuity design, Fixed-effect regression, and matching techniques including Propensity score matching and Entropy balancing
 - Machine Learning (Regularization methods, KNN, Decision Tree ensemble methods)
- ⇒ Qualitative Data Analysis: Applied Thematic analysis, Content analysis
- ⇒ LaTeX, a software program for typesetting mathematical expressions, and Wolfram Mathematica 8, a computational software program (Both applied for master's thesis)
- ⇒ Microsoft Office applications (Word, Excel, PowerPoint)
- ⇒ Geogebra (an interactive geometry, algebra, statistics and calculus application)
- ⇒ Artificial Intelligence applications in education

TEACHING EXPERIENCE (SOME COURSES TAUGHT)

- MIG507 Statistical Data Analysis (Graduate-level course in the Institute for Immigration Studies, taught in English)
- Algorithm and Programming
- Computer-Assisted Mathematics Instruction
- Problem Solving in Mathematics
- Mathematics Teaching and Learning Methods
- Middle School Mathematics Curriculum
- Inclusive Practices in Teaching Mathematics
- Assessment in Mathematics Classrooms
- Connections in Teaching Mathematics
- Communication in Mathematics Classrooms
- Vocational English
- Research Techniques
- Business Mathematics (Vocational College level)
- Vocational Mathematics (Vocational College level)
- Vocational English I & II (Vocational College level)
- Vocational English Applications (Vocational College level)

SERVICES & COMMUNITY OUTREACH

Establishing a Mathematics Library in Nizip College of Education (TDP Project), 2026 [Link](#)

Organizing a visit to Innovative Approaches in Mathematics Education Exhibition, 2025 [Link](#)

University Students Volunteer Along with Their Younger Siblings (TDP Project: Fundraising Event for students in poverty), 2024 [Link](#)

University Students Meet Their Younger Siblings (TDP Project: Educational and recreational activities and delivering aid packages for primary school students in a poverty-stricken area), 2022 [Link](#)

HOBBIES

Reading, Playing Billiards, Fishing, Traveling, Jogging, Hiking

[Last Updated: July 24, 2026]